



CSEE 4290: Computer Architecture

Top Level SCC – Spring 2025 – Group 5

[GitHub](#)

Completed

Instruction Fetch (No Branching) – Ricky Chen

Instruction Decode (Just Decode) – Matthew Sherritt

Register File – Ricky Chen

To-Do

IF Unconditional Branch – Ricky Chen

ID (Fully Coded) – Matthew Sherritt

EXE Start – Nathaniel LeBlanc

Group Responsibilities

Test Bench – Nathaniel LeBlanc

Instruction Fetch – Ricky Chen

Instruction Decode – Matthew Sherritt

Execute – Nathaniel LeBlanc

Memory Module – Nathaniel LeBlanc

Register File – Ricky Chen

Schedule

IF/ID – Oct. 3

Execute 3 Instructions/Branch Imm – Oct. 10

Execute Complete – Oct. 17

Testcases – Oct. 24

SCC Complete – Oct. 29

Top Level I/O

```
module scc_f25_top(clk,rst,clk_en,halt_f,err_bits,instruction_memory_v,data_memory_in_v);  
  
    input  clk,rst,clk_en;  
    output halt_f;  
    output [1:0] err_bits;  
    output [31:0] instruction_memory_v,data_memory_in_v;  
  
    wire instruction_memory_en, data_memory_read, data_memory_write;  
    wire [31:0] instruction_memory_a, data_memory_a, data_memory_out_v;
```